

SIF SENTINEL

AI-Powered SIF Precursor Intelligence & Petrobras 3W Oil-Well Event Intelligence Platform

Problem Statement Alignment: SIH26165 | **Tagline:** 'Don't wait for the incident. Find the precursor.'

Technical Report Date: August 27, 2026 | **Version:** 2.0.0 (Hardened Production Readiness)

Core Product Positioning & Responsible AI Commitment:

SIF Sentinel is an explainable safety intelligence system that connects differently worded safety observations to uncover recurring Serious Injury & Fatality (SIF) precursor patterns, identify repeatedly failing preventive barriers, detect emerging risk, and support human-led preventive action. **SIF Sentinel does NOT claim to predict worker fatalities or exact industrial accidents.** It empowers human safety engineers with grounded precursor intelligence.

1. System Architecture & Closed-Loop Intelligence Pipeline

SIF Sentinel operates on a 9-stage closed-loop safety paradigm: **REPORT** → **UNDERSTAND** → **CONNECT** → **IDENTIFY** → **WARN** → **VALIDATE** → **ACT** → **MEASURE** → **IMPROVE**. The architecture unifies deterministic safety ontology extraction, dense semantic sentence embeddings, density-based clustering, multi-factor risk quantification, and human-in-the-loop expert review.

Layer	Technology / Component	Exact Version	Role & Architecture
Frontend	Next.js App Router, React, Tailwind CSS	16.3.2 / 19.2.8	Interactive Command Center, Graphs, 3W Time-Series Visualizer
Backend	FastAPI, Uvicorn, Pydantic	0.111.0 / 2.7.4	Asynchronous REST API, Ingestion, Scoring, Inference
NLP / ML	Sentence-Transformers (all-MiniLM-L6-v2)	3.0.1 (384-dim)	Dense semantic vectorization of safety report narratives
Clustering	Scikit-Learn DBSCAN & Cosine Distance	1.9.0 (Pinned)	Density clustering (eps=0.45, min_samples=2) with noise isolation
Database	SQLAlchemy ORM (SQLite / PostgreSQL)	2.0.31 / pgvector	Zero-config local SQLite + Production PostgreSQL pgvector schema
3W Module	Random Forest with class_weight='balanced'	100 Trees, Depth 12	Time-series operational event classifier on Petrobras 3W 2.0.0

2. Empirical NLP & Safety Precursor Extraction Evaluation

SIF Sentinel integrates a domain ontology covering 9 critical industrial hazard domains (Electrical, Working at Height, Confined Space, Process Safety, Lifting/Rigging, Chemical, Excavation, Line of Fire, Vehicles). It features regex-bounded negation handling ensuring compliant procedures (e.g. 'LOTO was followed') are never misclassified as failures, while active omissions (e.g. 'LOTO was not followed') are correctly identified.

Evaluation Track	Dataset / Sample	Precision	Recall	F1 Score	Accuracy / Delta
Ontology (Dev Set)	46 Structured Samples	96.00%	66.67%	78.69%	73.9% Acc
Ontology (Held-Out)	50 Independent Samples	68.42%	37.14%	48.15%	52.0% Acc

Embeddings Space	all-MiniLM-L6-v2 (384-d)	Sim: 0.4161	Dissim: 0.0722	+0.3439 Margin	5.77x Contrast Ratio
DBSCAN Clustering	50 Held-Out Precursors	8 Clusters	62.0% Clustered	0.8425 Coherence	38.0% Noise Ratio
SIF Risk Separation	5-Factor Risk Formula	High: 56.6/100	Low: 26.0/100	+30.6 pts Delta	Distinct Separation

3. Multi-Dataset Integration & Data Provenance Separation

SIF Sentinel connects four distinct, validated public and industrial safety datasets without data contamination. Each dataset is assigned explicit metadata provenance:

Dataset Source	Volume / Format	Analytical Scope	Provenance Label
IHM Stefanini	425 CSV Records	Real-world industrial incident NLP extraction, potential severity vs actual severity preservation	IHM Stefanini — Public Industrial Safety Dataset
OISD Case Studies	92 PDF Documents	Indian Oil Industry Safety Directorate case studies, barrier failures & recommendations	OISD — Indian Oil & Gas Safety Publications
BSEE Investigations	2,016 CSV Records	Bureau of Safety & Environmental Enforcement GOM offshore incident frequency & recurrence trends	BSEE — Offshore Incident Investigation Data
Petrobras 3W 2.0.0	2,228 Parquet Files (1.74 GB)	Multi-sensor oil-well telemetry for operational event classification (10 classes)	Petrobras 3W Dataset 2.0.0 — Oil-Well Time-Series

4. Petrobras 3W Oil-Well Operational Event Intelligence Module

The 3W module is an independent time-series machine learning component built to classify 10 official operational event states from multi-sensor well telemetry. It utilizes a streaming lazy loader (preventing RAM exhaustion), extracts 45 domain-grounded sensor features (pressures, temperatures, choke ratios, onset deltas), and trains a balanced Random Forest model.

Held-Out Test Set Performance (442 Independent Instances, Zero Leakage):

Metric	Score	Benchmark Baseline	Performance Significance
Macro F1 Score	98.93%	4.21% (Majority Class)	+2,247.6% Lift over trivial classification across 10 classes
Balanced Accuracy	99.36%	10.00% (Random Guess)	Proves strong sensitivity across rare classes (DHSV closure, PCK scaling)
Weighted F1 Score	99.10%	11.23% (Majority Class)	Robust generalizability across full instance distribution
Raw Accuracy	99.10%	26.70% (Class 0 Baseline)	438 / 442 test instances correctly classified

3W Per-Class Performance Breakdown (10 Official Classes):

CID	Official Event Class Name	Precision	Recall	F1 Score	Test Support
0	Normal Operation	100.00%	98.31%	99.15%	118
1	Abrupt Increase of BSW	100.00%	100.00%	100.00%	25
2	Spurious Closure of DHSV	100.00%	100.00%	100.00%	7
3	Severe Slugging	100.00%	100.00%	100.00%	21
4	Flow Instability	95.77%	100.00%	97.84%	68
5	Rapid Productivity Loss	100.00%	100.00%	100.00%	90
6	Quick Restriction in PCK	100.00%	97.73%	98.85%	44
7	Scaling in PCK	90.00%	100.00%	94.74%	9
8	Hydrate in Production Line	100.00%	100.00%	100.00%	19
9	Hydrate in Service Line	100.00%	97.56%	98.77%	41

5. Platform Capabilities & Human-in-the-Loop Safeguards

SIF Sentinel integrates advanced decision-support features ensuring safety intelligence leads to tangible risk reduction:

Feature Area	Technical Mechanism	Operational Value
Barrier Health	Degradation formula based on precursor velocity, multi-site spread & severity. Persists historical snapshots.	Identifies crumbling physical & procedural controls before severe barrier breaches occur.
Connect the Dots	Multi-node graph linking reports across sites, dates, equipment, and shared failure modes.	Reveals cross-facility latent systemic hazards invisible in siloed site spreadsheets.
Expert Review	Human-in-the-loop review interface allowing safety officers to validate, override, or escalate AI findings.	Ensures algorithmic suggestions are strictly vetted by certified HSE engineers.
Preventive Actions	Closed-loop tracking with pre/post intervention recurrence velocity measurement.	Provides quantifiable audit proof of risk reduction following corrective actions.
What-If Simulator	Synthetic barrier intervention simulation modeling projected SIF score reductions.	Supports data-driven budget and maintenance resource prioritization.
Safety Copilot	Grounded natural language telemetry interface citing live database evidence without hallucination.	Answers executive queries ("Which barrier is degrading fastest?") with built-in culture safeguards.

6. Quality Assurance, Security Audit & Build Verification

Automated Test Suite: 25 / 25 automated tests passed in 25.87s across core NLP, risk mathematics, 3W ML pipeline, OISD parser, and BSEE analytics.

Frontend Production Build: Next.js 16 (Turbopack) production build completed cleanly in 14.7s with 0 TypeScript errors across all 13 routes.

Security & Privacy Audit: Scanned codebase for hardcoded credentials, API keys, and machine paths. Verified 0 exposed secrets.